

Pushkar Ambedpelliwar

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EDUCATION

Georgia Institute of Technology

B.S. Mechanical Engineering, Robotics & Automation Concentration

Atlanta, GA

Expected May 2028

- **GPA:** 3.64/4.00
- **Relevant Coursework:** Statics, Dynamics of Rigid Bodies, Deformable Bodies, Thermodynamics, Fluid Mechanics, System Dynamics, Circuits & Electronics, Computing for Engineers, Data Structures & Algorithms, Object-Oriented Programming

EXPERIENCE

Undergraduate Student Researcher

GT Dong Lab

April 2025 – May 2026

Atlanta, GA

- Built a MATLAB pipeline converting 3D Slicer aortic CT scans into patient-specific FEM meshes using threshold-based segmentation and automated meshing
- Ran 250+ structural FEA analyses on the generated meshes under varying load conditions to produce training data for an ML rupture-prediction model, achieving 82% accuracy

Mechanical Design and Manufacturing Engineer

Industry Innovation, Circular Economy Lab

August 2025 – May 2026

Atlanta, GA

- Optimized a customizable PLA bottle design in SolidWorks for manufacturability, applying GD&T and DFM-driven wall thickness and draft angle changes to cut material waste 8% and reduce scrap rate
- Managed injection molding production for 14 custom mechanical components, tuning preform geometry and tooling to hold dimensional tolerances across production runs
- Automated SolidWorks parametric updates with a MATLAB script parsing Excel into the Global Variables Library, eliminating manual geometry re-entry across design iterations

Student Researcher – Biomechanical Applications

CSU Prawel Lab

April 2024 – August 2024

Fort Collins, CO

- Fabricated 15 patient-specific biomimetic scaffolds via 3D printing and dynamic cell seeding; wrote MATLAB code to simulate strain distributions under 4 kph walking loads
- Rendered 10 patient bone models from μ CT scans in InVesalius and modified geometry in SolidWorks for surgical implantation planning

PROJECTS

Autonomous Indoor Mapping Rover | SolidWorks, KiCAD, ROS 2, C++, Python | GitHub

May 2026 – Present

- Fabricated a 10-standoff tiered rover chassis in SolidWorks, validating the 3mm plywood tier design via hand-calculated beam analysis (29x safety factor) before build
- Corrected real URDF/mesh alignment errors (wheel geometry, caster offset) against SolidWorks-measured ground truth, establishing an assembly-relative STL export and revision workflow
- Ran a closed-loop 10-trial repeated-rotation test and performed root-cause failure analysis to isolate a 10.3% systematic heading bias in an IMU/EKF sensor fusion pipeline, distinguishing a true system fault from random noise via trial-to-trial variance
- Designed a KiCAD schematic and wired servos and an ultrasonic sensor over UART into the rover's electromechanical system, coordinating mechanical and electrical design

Robotic Arm and Control System | Fusion 360, Arduino, Java

January 2025 – May 2025

- Modeled a 6-DOF robotic arm in Fusion 360, running a polymer-vs-metal material trade study to cut link inertia and reduce peak motor torque
- Engineered a Java motor control system integrated with an Arduino haptic sensor suite for closed-loop grasp control

Biorobotic System Integration | Autodesk Eagle, SolidWorks, MATLAB, Arduino

January 2025 – July 2025

- Designed and soldered a custom PCB and 3D-printed a mounting harness in SolidWorks to acquire 10 ms-resolution EMG signals from a live specimen
- Built a MATLAB bandpass filtering pipeline to remove noise and isolate the muscle activation frequency band, extracting onset timing from raw signal data

TECHNICAL SKILLS

CAD/CAE: SolidWorks, Fusion 360, FEA, Hand Calculations, DFM, GD&T

Manufacturing: Injection molding, 3D printing, tolerancing, tooling, process automation

Programming: C++, C, Python, MATLAB, Java

Electromechanical: PCB Layout (Autodesk Eagle, KiCAD), Circuits & Electronics, UART/SPI/I2C

Robotics: ROS 2 Jazzy, Nav2, SLAM Toolbox, Arduino

Tools: Docker, Git, Linux (Ubuntu 24.04)

Certifications: Certified SolidWorks Professional (CSWP)